Training 2: Simple examples of structural optimization

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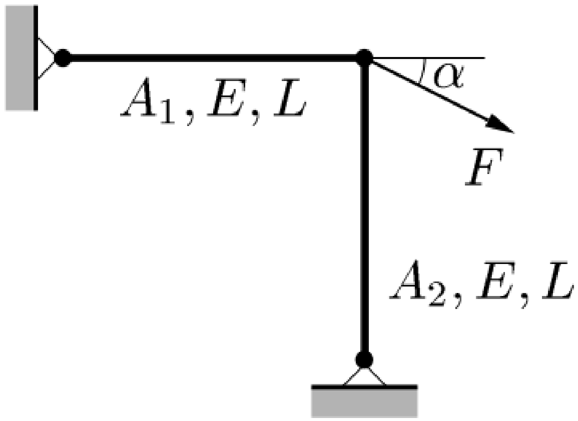
Exercise 1



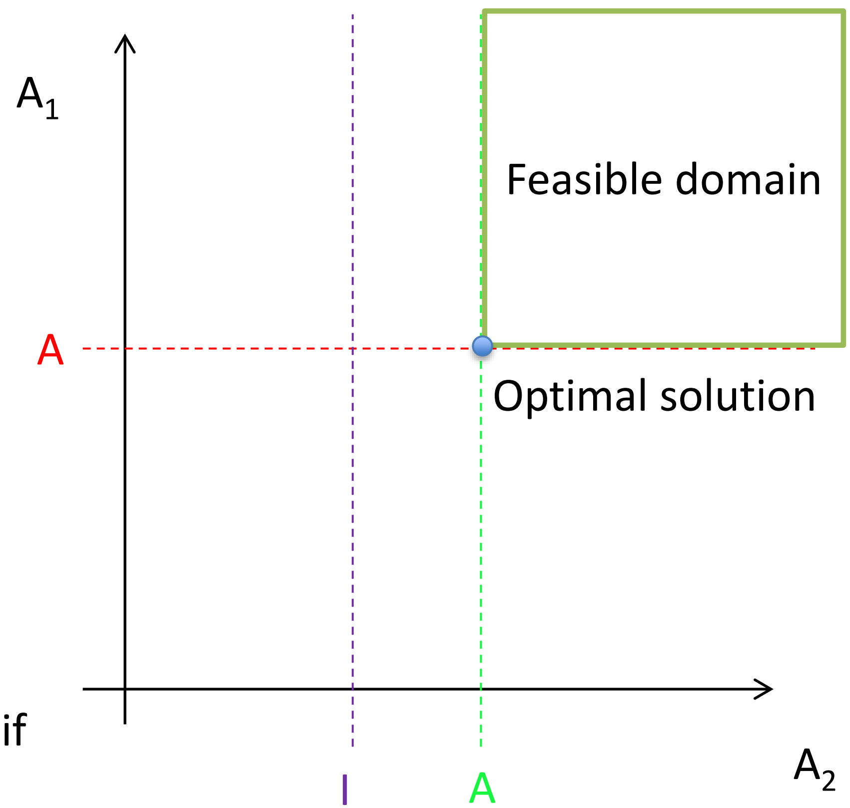
Consider a two-bar truss consisting of bars of length L and Young's modulus E. The force F > 0 is applied at an angle α = 45°.

The problem is **to find the circular cross-sectional areas A_1 and A_2 such that the weight of the truss is minimized under constraints on stresses and Euler buckling (with a safety factor of 4)**.

$$(\mathbb{SO}) \quad \begin{cases} \text{minimize } f(x, y) \text{ with respect to } x \text{ and } y \\ \text{subject to} \quad \begin{cases} \text{behavioral constraints on } y \\ \text{design constraints on } x \\ \text{equilibrium constraint.} \end{cases} \end{cases}$$

Exercise 1

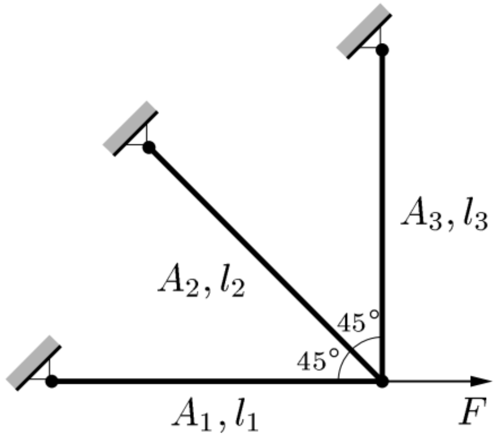
$$(\mathbb{SO})^2_{\text{nf}} \quad \begin{cases} \min\limits_{A_1, A_2} & A_1 + A_2 \\ \text{s.t.} & \begin{cases} A_1 \geq \dfrac{F}{\sqrt{2}\sigma_0} \\ A_2 \geq \dfrac{F}{\sqrt{2}\sigma_0} \\ A_2^2 \geq \dfrac{16FL^2}{\sqrt{2}\pi E}. \end{cases} \end{cases}$$



A = Active constraint
I = Inactive constraint

An *active* constraint is a constraint that, if removed, changes the optimal solution.

Exercise 2



Consider the three-bar truss shown.
The bars have Young's modulus E and the lengths are $l_1 = L$, $l_2 = L$, $l_3 = L/\beta$, where $\beta > 0$. The force $F > 0$.
We are to minimize the weight under stress constraints. The **design variables are** the cross-sectional areas $\mathbf{A_1}$**,** $\mathbf{A_2}$ and $\mathbf{A_3}$.

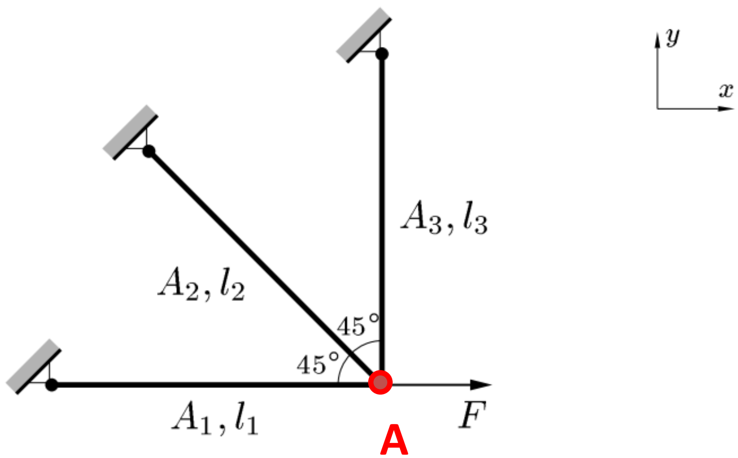
In order to simplify some calculations, we introduce the following assumption:

$$A_1 = A_3$$

Therefore, the mass of the structure (objective function) can be written as:

$$f(A_1, A_2) = \rho_1 L A_1 + \rho_2 L A_2 + \rho_3 \frac{L}{\beta} A_3 = L\left(\rho_1 + \frac{\rho_3}{\beta}\right) A_1 + L\rho_2 A_2,$$

Exercise 2



Again, the aim is to express all the constraints as function of the design variables, to write the problem using the nested formulation.

The design constraint is:

$$A_1 \geq 0, \qquad A_2 \geq 0.$$

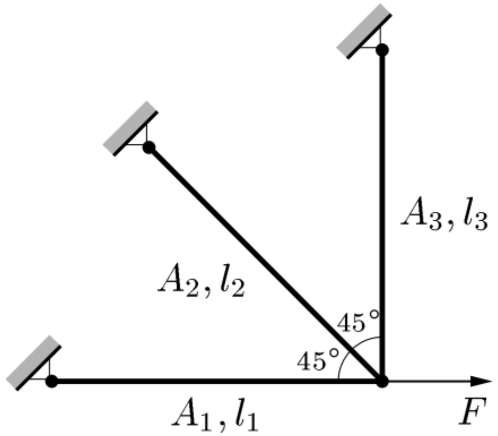
The state constraint is:

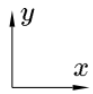
$$|\sigma_i| \leq \sigma_i^{\max}, \qquad i = 1,\ 2,\ 3.$$

Writing the equilibrium of forces in node **A**, one gets:

$$-s_1 - \frac{s_2}{\sqrt{2}} + F = 0, \qquad s_3 + \frac{s_2}{\sqrt{2}} = 0. \quad \rightarrow \begin{bmatrix} F \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{\sqrt{2}} & 0 \\ 0 & -\frac{1}{\sqrt{2}} & -1 \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \iff \boldsymbol{F} = \boldsymbol{B}^T \boldsymbol{s}.$$

Exercise 2





The system is statically indeterminate, therefore, in order to solve it, one needs to **account for its deformability**.
To do that, it is necessary to link the stress state to the deformation.

$$s_i = \frac{E A_i \delta_i}{l_i}.$$

How to find the equation:

Linear elastic material +
De Saint Venant's theory

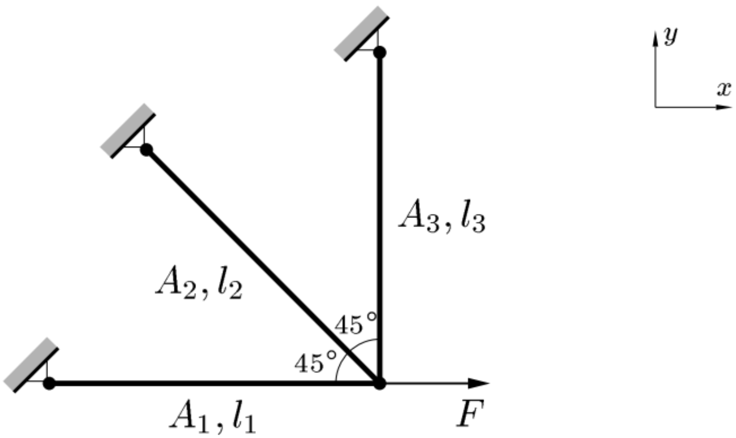
$$\sigma = \frac{N}{A}$$

$$\sigma = E\varepsilon$$

$$\varepsilon = \frac{\delta}{L}$$

$$N = \sigma A = EA\varepsilon = EA\frac{\delta}{L}$$

Exercise 2



Now, introducing some matrices:

$$\boldsymbol{s} = \boldsymbol{D}\boldsymbol{\delta}, \qquad \boldsymbol{D} = \frac{E}{l}\begin{bmatrix} A_1 & 0 & 0 \\ 0 & A_2 & 0 \\ 0 & 0 & \beta A_1 \end{bmatrix}$$

Since, from theory, δ = Bu:

$$\boldsymbol{s} = \boldsymbol{D}\boldsymbol{B}\boldsymbol{u}.$$

Therefore:

$$\boldsymbol{F} = \boldsymbol{B}^T\boldsymbol{s} = \boxed{\boldsymbol{B}^T\boldsymbol{D}\boldsymbol{B}}\boldsymbol{u} = \boldsymbol{K}\boldsymbol{u},$$

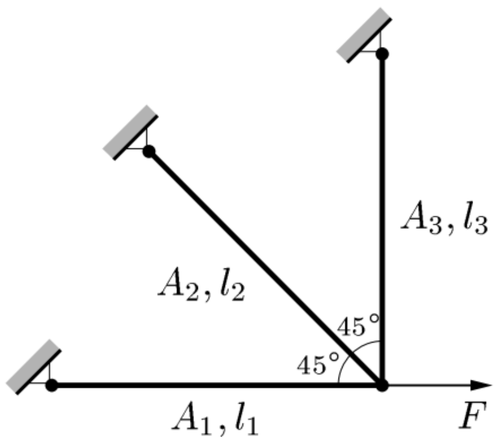
$$\downarrow$$

$$\mathbf{k} = \int \mathbf{B}^T\mathbf{E}\mathbf{B}\,dV$$

$$\boldsymbol{K} = \frac{E}{l}\begin{bmatrix} A_1 + \dfrac{A_2}{2} & -\dfrac{A_2}{2} \\ -\dfrac{A_2}{2} & \dfrac{A_2}{2} + \beta A_1 \end{bmatrix}.$$

K is called **global stiffness matrix**.

Exercise 2



Eventually, stresses are obtained:

$$\boldsymbol{\sigma} = \boldsymbol{A}\boldsymbol{s} = \boldsymbol{A}\boldsymbol{D}\boldsymbol{B}\boldsymbol{u},$$

where

$$\boldsymbol{A} = \begin{bmatrix} \frac{1}{A_1} & 0 & 0 \\ 0 & \frac{1}{A_2} & 0 \\ 0 & 0 & \frac{1}{A_1} \end{bmatrix}.$$

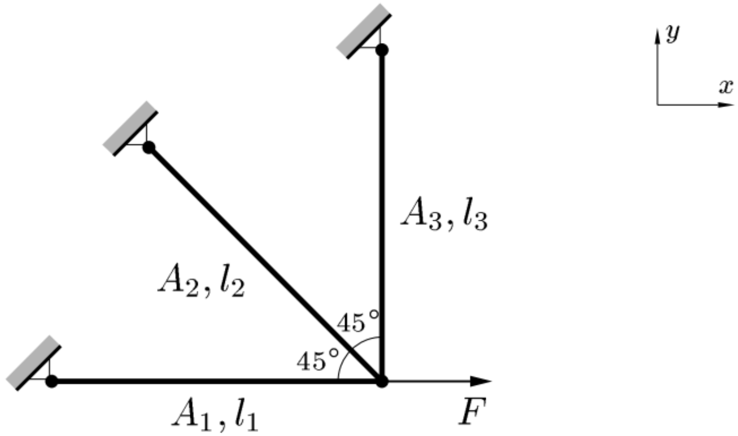
To recap:

Bu gives the elongations δ, that is, the matrix **B** links the nodal displacements u to the elongations δ.

Then, **DB**u, gives the internal forces inside the elements, that is, **D** transforms elongations (close to strains) to forces (close to stresses). **D** is a matrix that depends on the material.

Eventually, **ADB**u results in stresses. Indeed matrix **A** is just a diagonal matrix which divides internal forces s for areas A, to get stresses.

Exercise 2



Eventually, **scalar** formulae for stresses are obtained:

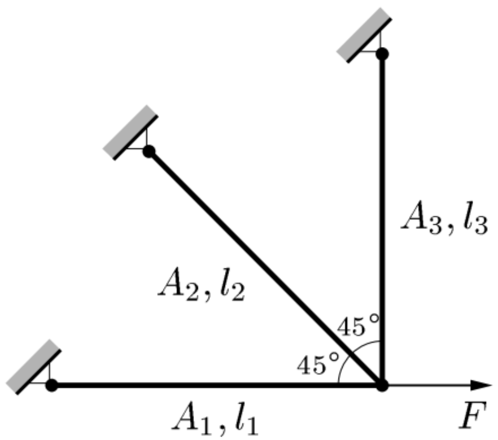
$$\sigma_1 = \frac{F}{2\beta A_1 + (1+\beta)A_2}\left(2\beta + \frac{A_2}{A_1}\right),$$

$$\sigma_2 = \frac{\sqrt{2}F\beta}{2\beta A_1 + (1+\beta)A_2},$$

$$\sigma_3 = -\frac{F\beta\dfrac{A_2}{A_1}}{2\beta A_1 + (1+\beta)A_2}.$$

Due to the constraint F > 0, **bars 1 and 2 are in tension**, while **bar 3 is in compression**.

Exercise 2



Introducing the hypothesis β = 1, **the constraints can be formulated**:

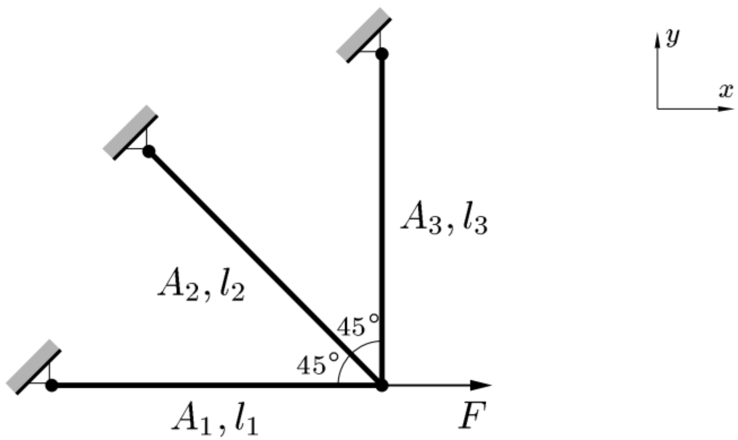
$$\frac{F(2A_1 + A_2)}{2A_1(A_1 + A_2)} \leq \sigma_1^{\max}$$

$$\frac{F}{\sqrt{2}(A_1 + A_2)} \leq \sigma_2^{\max}$$

$$\frac{FA_2}{2A_1(A_1 + A_2)} \leq \sigma_3^{\max}$$

Recall that β was linked to the length of bar 3: $l_3 = L/\beta$

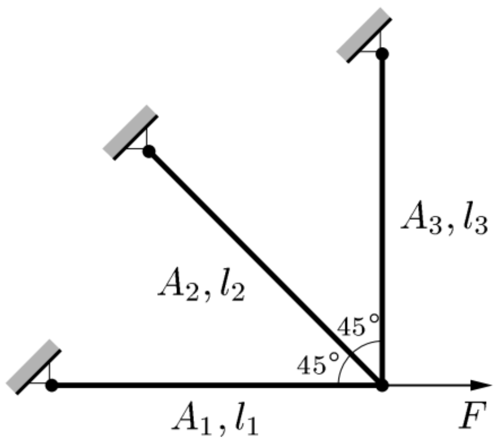
Exercise 2

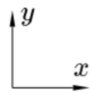


The final **nested formulation** is:

$$
(\mathbb{SO})^5_{\text{nf}} \quad \begin{cases} \min\limits_{A_1, A_2} & (\rho_1 A_1 + \rho_2 A_2 + \rho_3 A_1)\, L \\ \text{s.t.} & \begin{cases} \dfrac{F(2A_1 + A_2)}{2A_1(A_1 + A_2)} - \sigma_1^{\max} \leq 0 \\ \dfrac{F}{\sqrt{2}(A_1 + A_2)} - \sigma_2^{\max} \leq 0 \\ \dfrac{F A_2}{2A_1(A_1 + A_2)} - \sigma_3^{\max} \leq 0 \\ A_1 > 0, \qquad A_2 \geq 0. \end{cases} \end{cases}
$$

Exercise 2
case A





Now, with the help of Matlab, we will investigate some solutions to the problem.

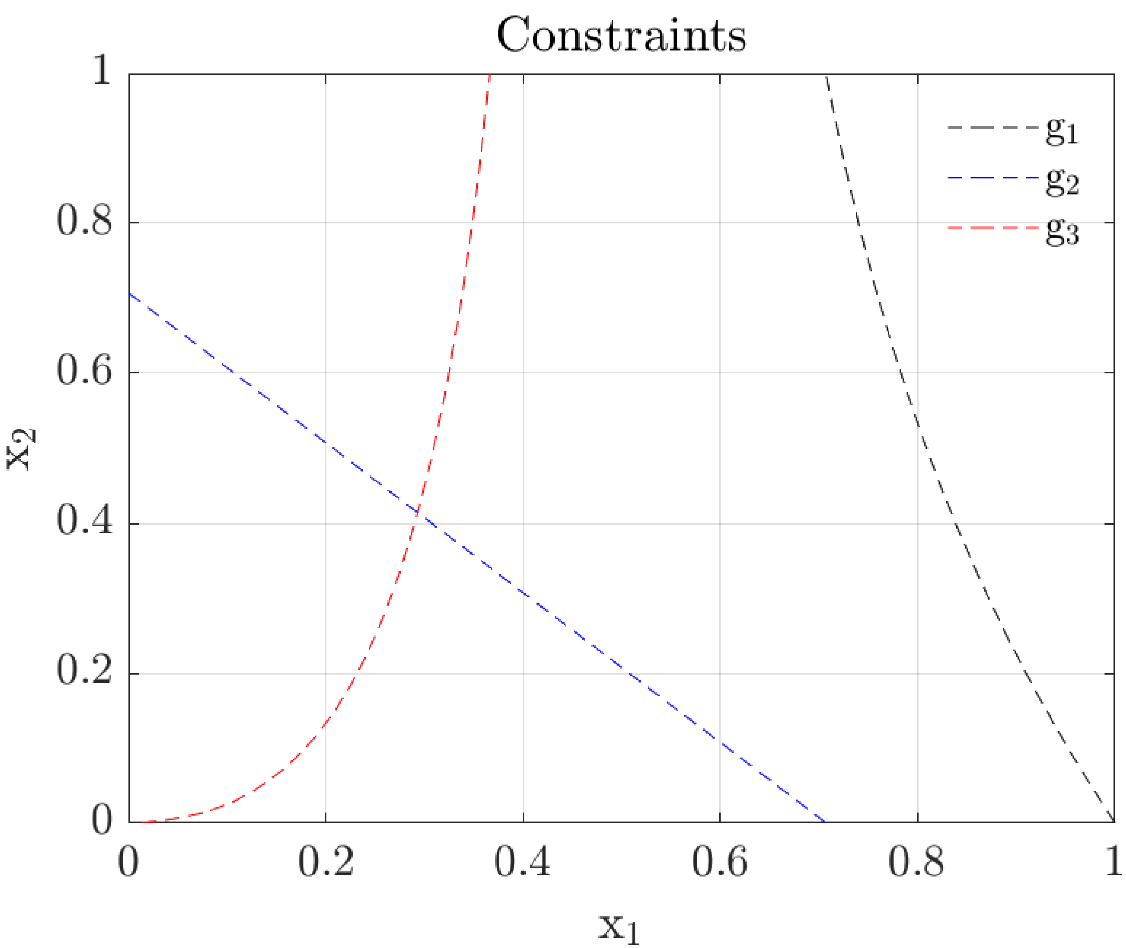
CASE A)
$\rho_1 = 2\rho_0,\ \rho_2 = \rho_3 = \rho_0,\ \sigma_1^{\max} = \sigma_2^{\max} = \sigma_3^{\max} = \sigma_0.$
By introducing the new dimensionless variables x_1 and x_2 as

$$x_1 = \frac{A_1\sigma_0}{F}, \qquad x_2 = \frac{A_2\sigma_0}{F},$$

Such that the problem becomes:

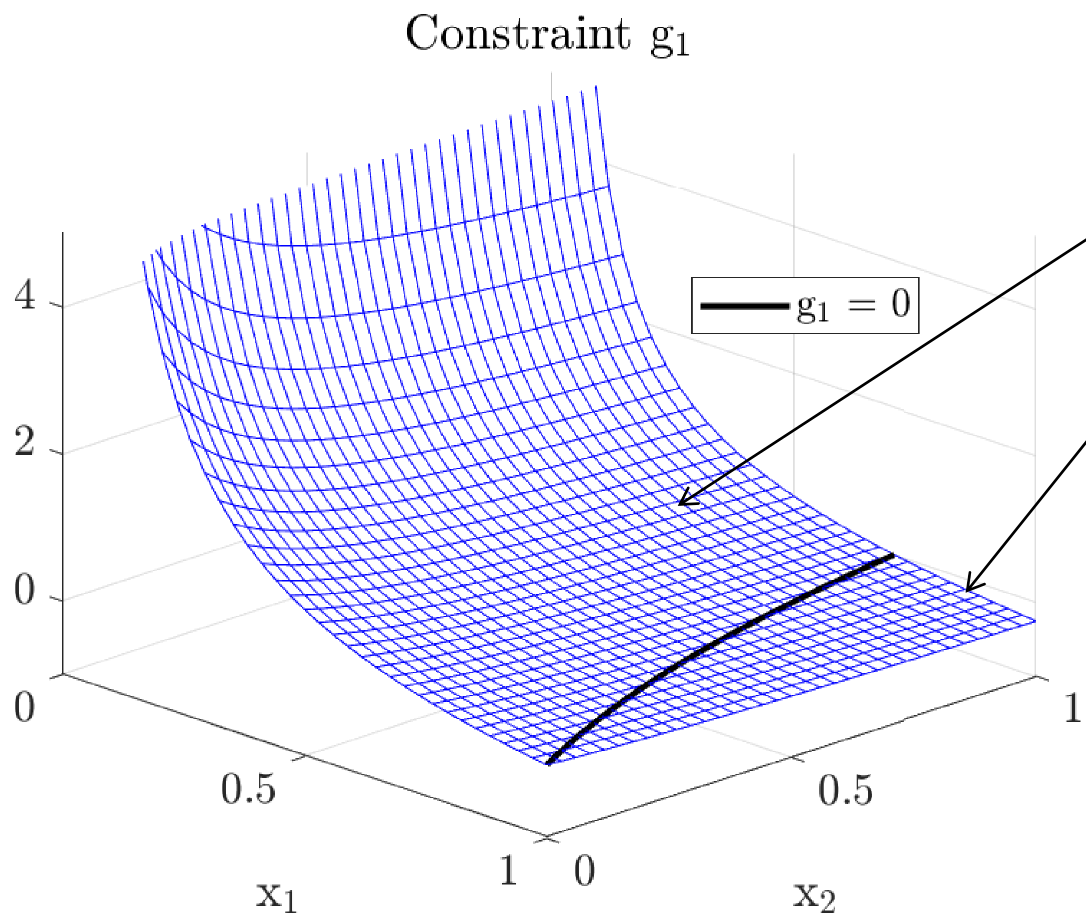
$$(\mathbb{SO})^{5a}_{\mathrm{nf}} \quad \begin{cases} \min\limits_{x_1, x_2} \ 3x_1 + x_2 \\ \text{s.t.} \quad \begin{cases} \dfrac{2x_1 + x_2}{2x_1(x_1 + x_2)} - 1 \leq 0 \\ \dfrac{1}{\sqrt{2}(x_1 + x_2)} - 1 \leq 0 \\ \dfrac{x_2}{2x_1(x_1 + x_2)} - 1 \leq 0 \\ x_1 > 0, \qquad x_2 \geq 0, \end{cases} \end{cases}$$

Exercise 2
case A



$$(\mathbb{SO})_{\mathrm{nf}}^{5a} \begin{cases} \min\limits_{x_1,x_2} & 3x_1 + x_2 \\ \text{s.t.} & \begin{cases} \dfrac{2x_1 + x_2}{2x_1(x_1+x_2)} - 1 \leq 0 \\ \dfrac{1}{\sqrt{2}(x_1+x_2)} - 1 \leq 0 \\ \dfrac{x_2}{2x_1(x_1+x_2)} - 1 \leq 0 \\ x_1 > 0, \qquad x_2 \geq 0, \end{cases} \end{cases}$$

Exercise 2
case A

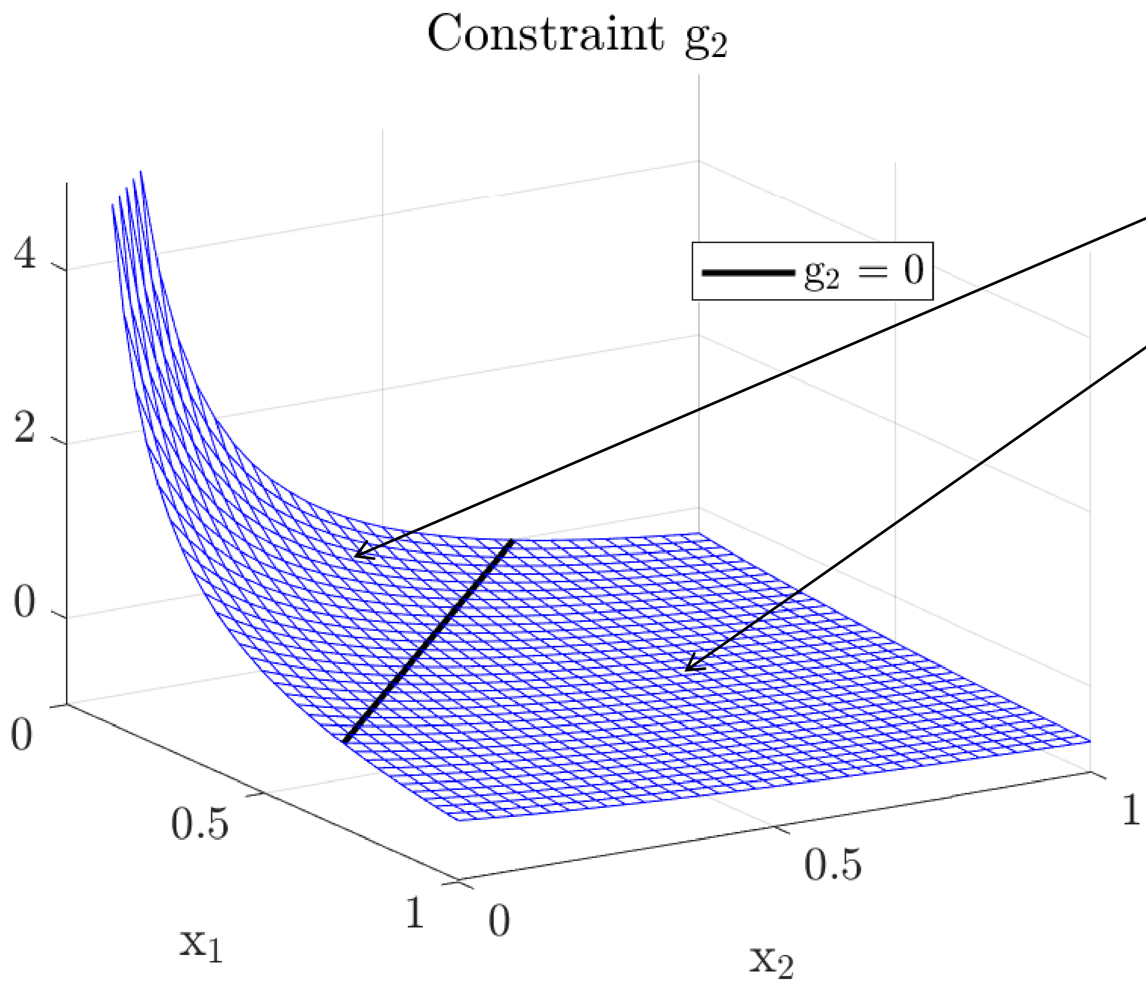


Unfeasible region

Feasible region

$$
(\mathbb{SO})^{5a}_{\text{nf}} \quad
\begin{cases}
\min\limits_{x_1, x_2} \;\; 3x_1 + x_2 \\[2ex]
\text{s.t.} \quad
\begin{cases}
\boxed{\dfrac{2x_1 + x_2}{2x_1(x_1 + x_2)} - 1} \leq 0 \\[2ex]
\dfrac{1}{\sqrt{2}(x_1 + x_2)} - 1 \leq 0 \\[2ex]
\dfrac{x_2}{2x_1(x_1 + x_2)} - 1 \leq 0 \\[1ex]
x_1 > 0, \qquad x_2 \geq 0,
\end{cases}
\end{cases}
$$

Exercise 2
case A

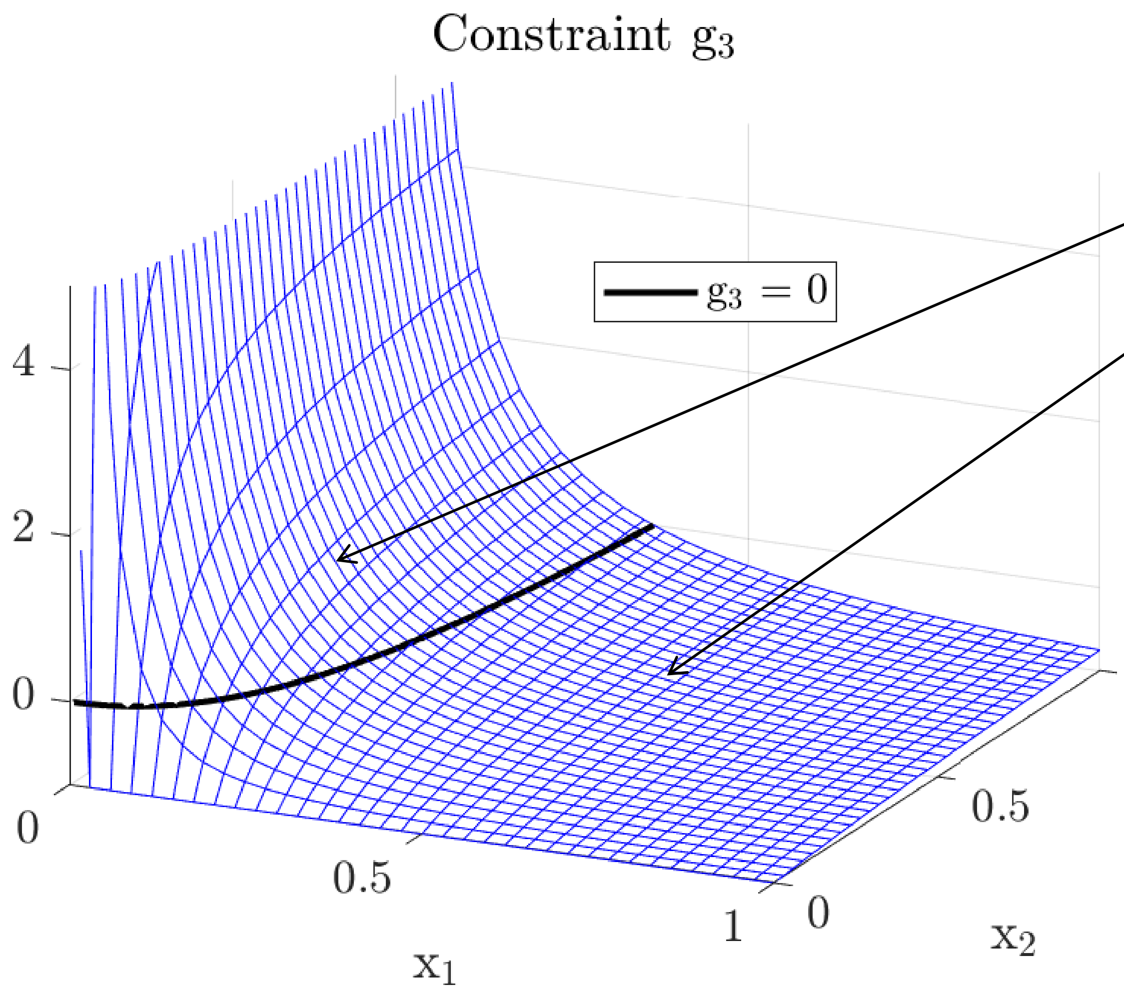


Unfeasible region

Feasible region

$$
(\mathbb{SO})^{5a}_{\text{nf}}
\begin{cases}
\min\limits_{x_1,x_2} & 3x_1 + x_2 \\
\text{s.t.} & \begin{cases}
\dfrac{2x_1 + x_2}{2x_1(x_1+x_2)} - 1 \le 0 \\
\boxed{\dfrac{1}{\sqrt{2}(x_1+x_2)} - 1} \le 0 \\
\dfrac{x_2}{2x_1(x_1+x_2)} - 1 \le 0 \\
x_1 > 0, \qquad x_2 \ge 0,
\end{cases}
\end{cases}
$$

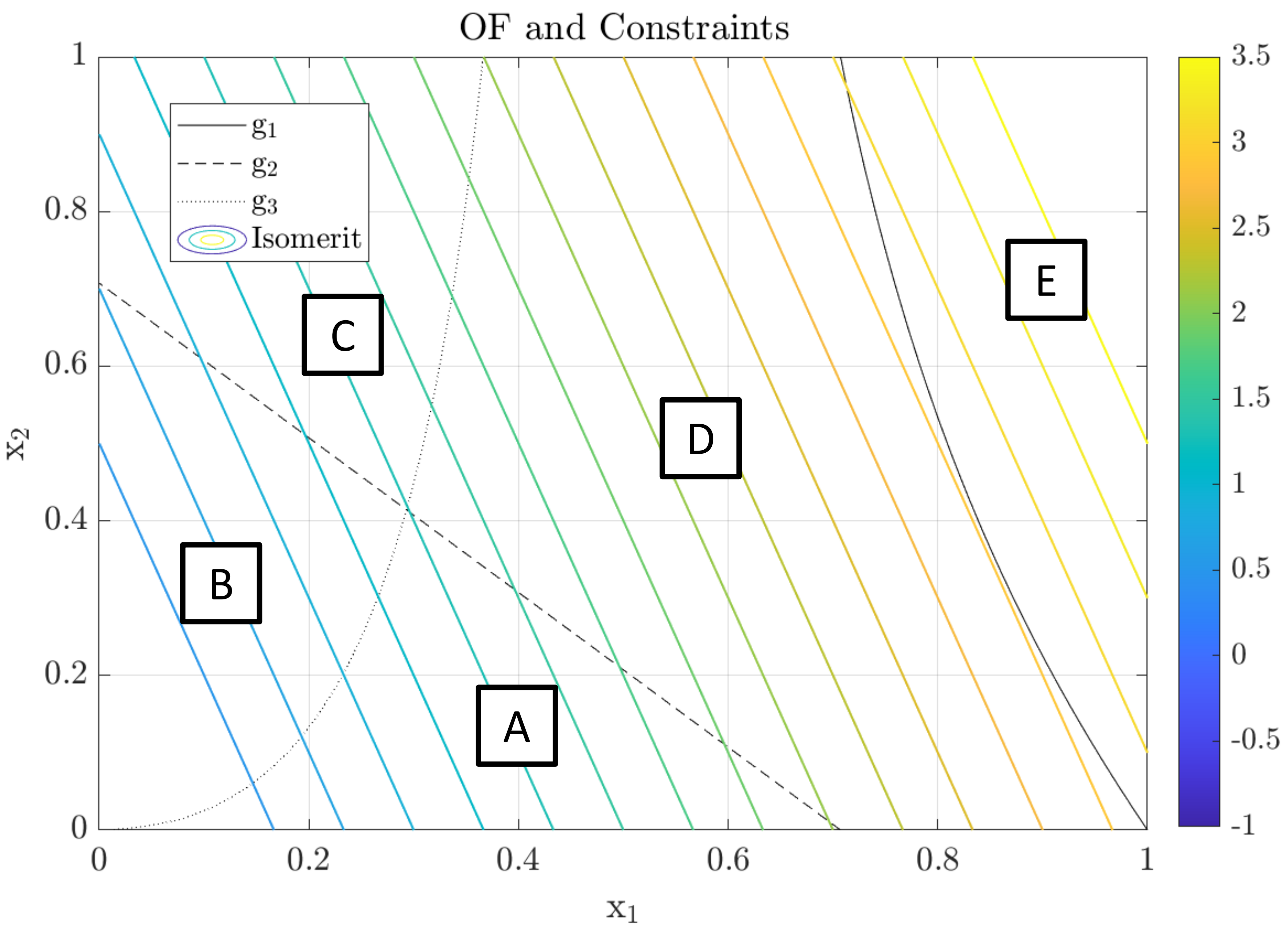
Exercise 2
case A



Unfeasible region
Feasible region

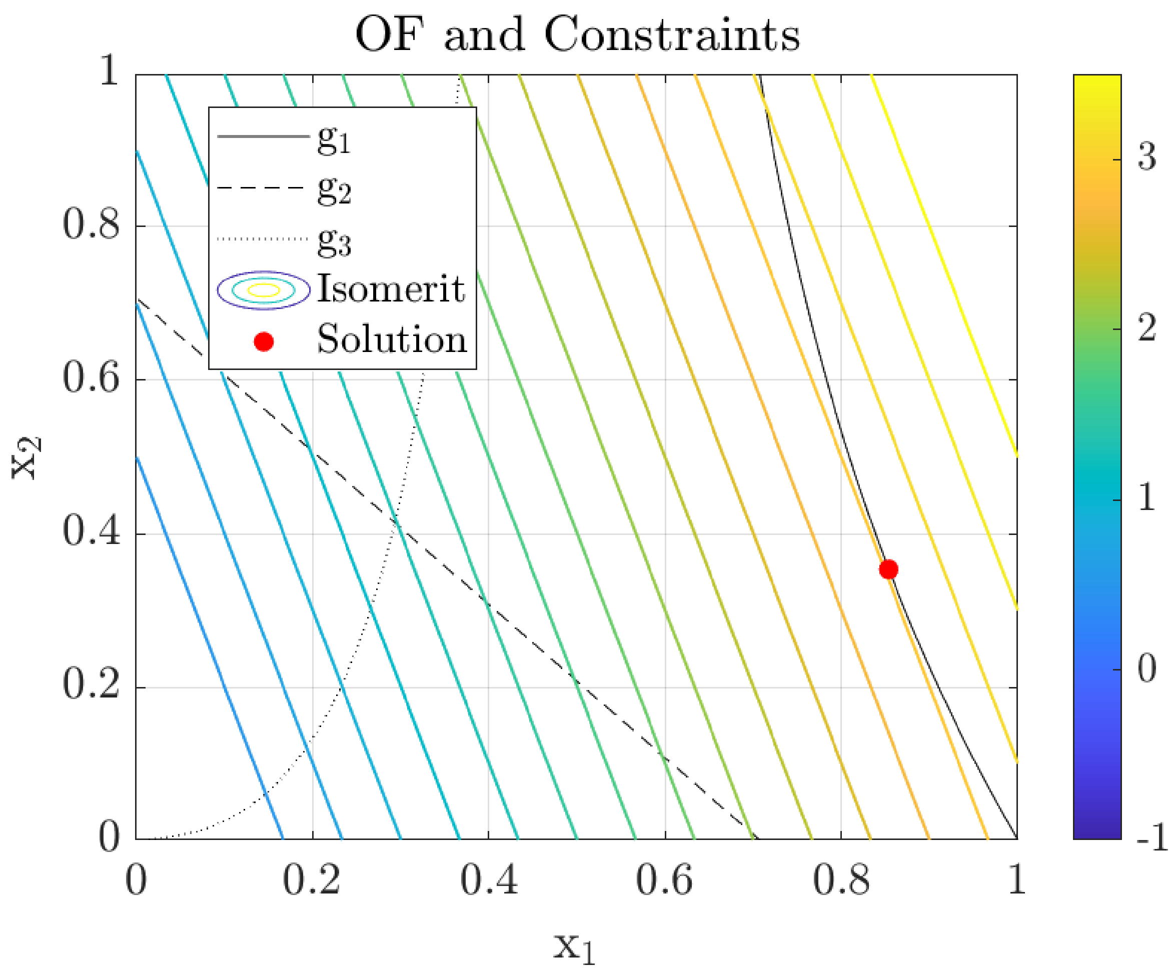
$$
(\mathbb{SO})^{5a}_{\mathrm{nf}} \quad
\begin{cases}
\displaystyle \min_{x_1,x_2} \; 3x_1 + x_2 \\
\text{s.t.} \quad
\begin{cases}
\dfrac{2x_1 + x_2}{2x_1(x_1+x_2)} - 1 \leq 0 \\
\dfrac{1}{\sqrt{2}(x_1+x_2)} - 1 \leq 0 \\
\boxed{\dfrac{x_2}{2x_1(x_1+x_2)} - 1} \leq 0 \\
x_1 \geq 0, \qquad x_2 \geq 0,
\end{cases}
\end{cases}
$$

Exercise 2
case A



Q. Which regions (A, B, C, D or E) do you think represent the feasible set? Why?

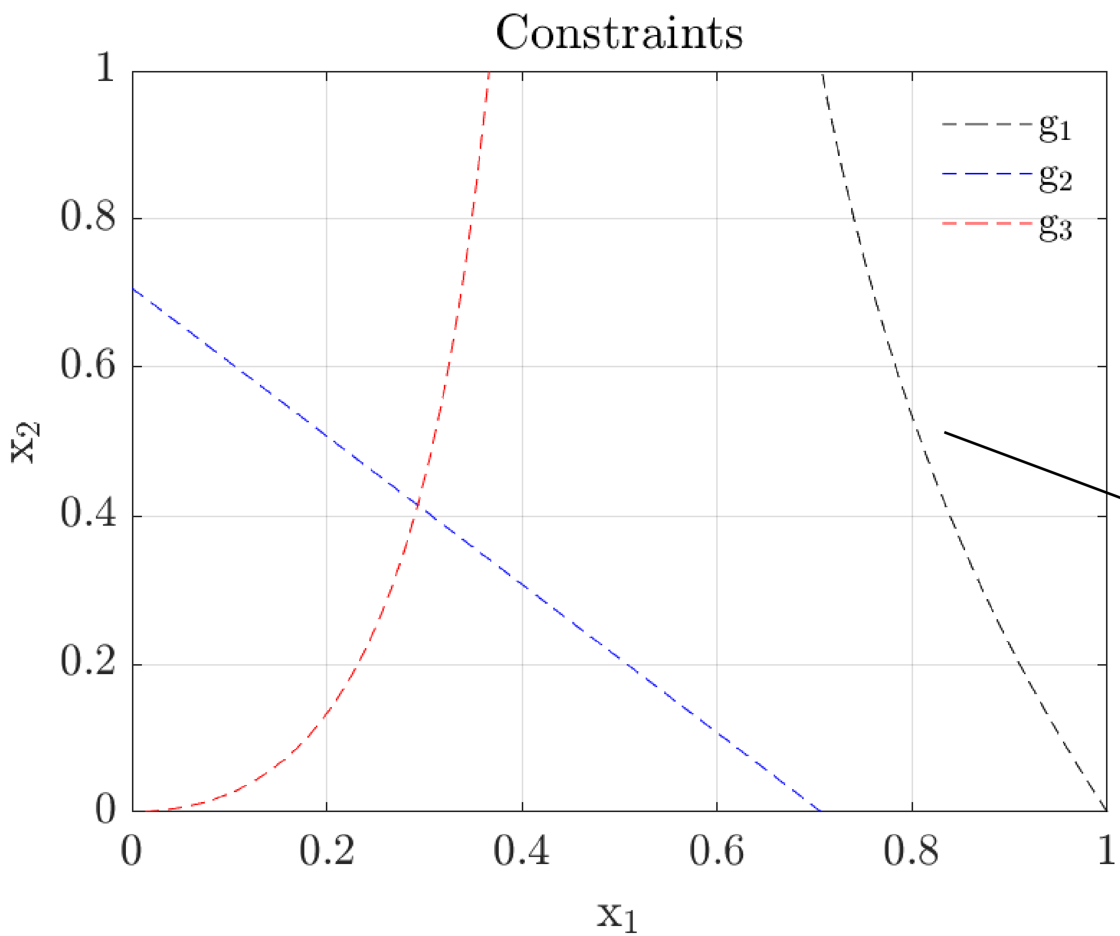
Exercise 2
case A



A. It is only region E, because it is the one which satisfies all constraints. The solution to the problem is the point which both belongs to the feasible set and minimizes the objective function value (i.e. lays on the lowest isomerit line).

Please note that no formal optimization criteria have been used to find the solution to this (simple) problem…

Exercise 2
case A



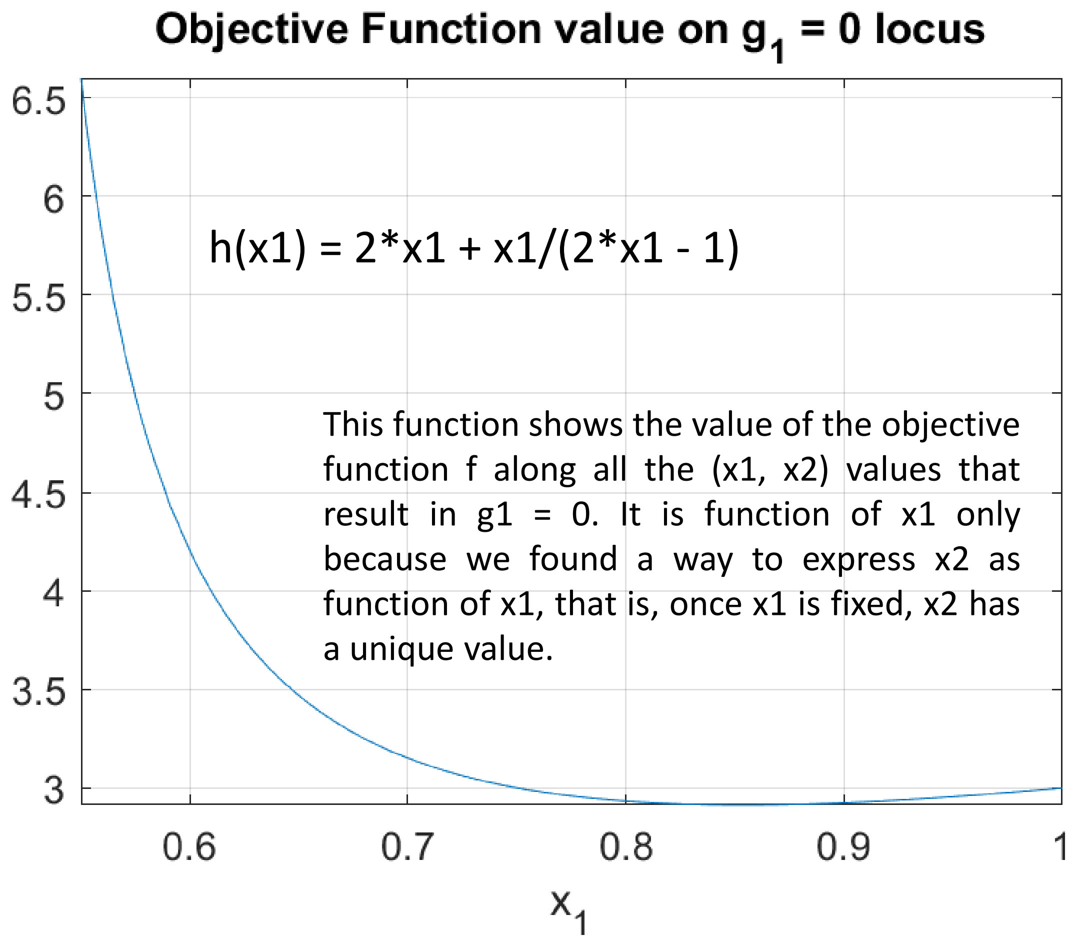
How to find the solution?

Once it is determined that g_1 is the active constraint, one should find the locus of (x1, x2) which produces $g_1 = 0$.

x2 = x1/(2*x1 - 1) - x1

(Found using Matlab Symbolic Toolbox)

Exercise 2
case A



Then, the locus x2(x1) is substituted inside the objective function f(x1, x2) to obtain a function of the variable x1 only:

Locus: x2(x1) = x1/(2*x1 - 1) - x1

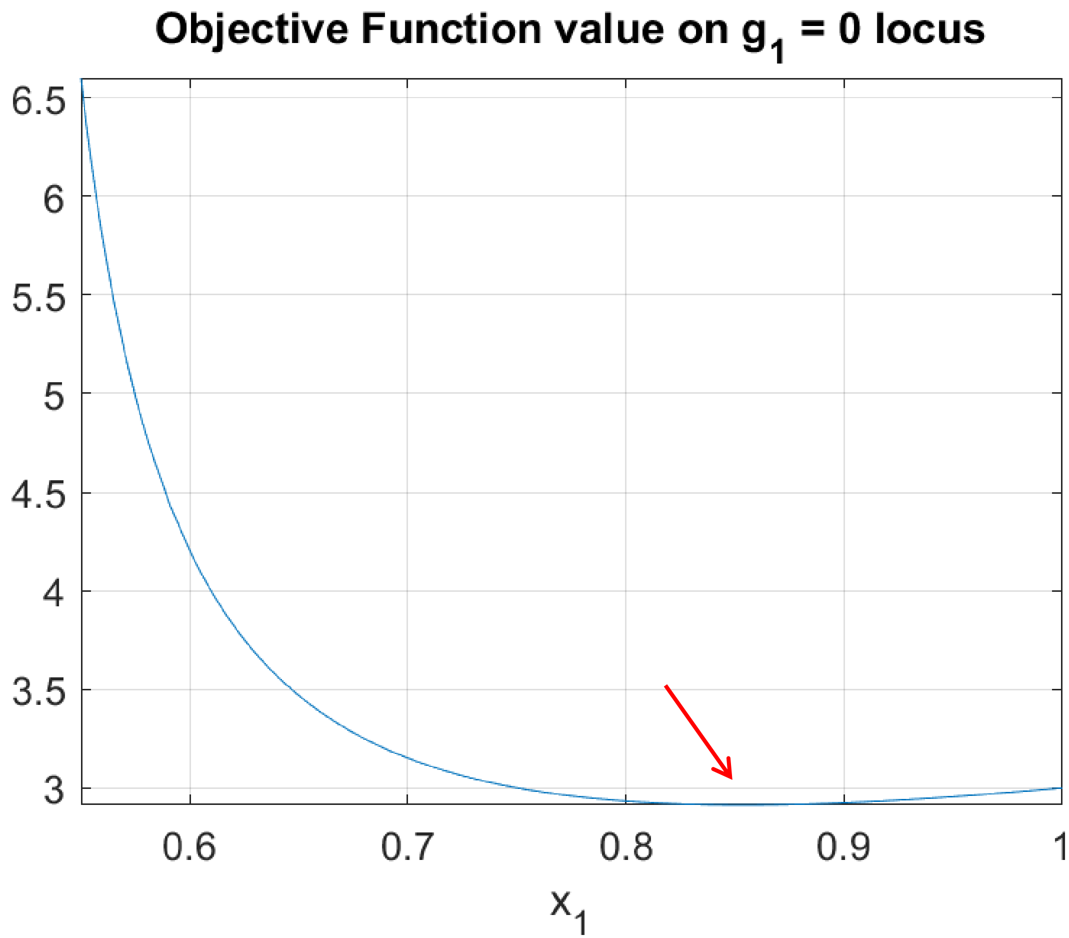
Substitute...

f(x1, x2) = 3*x1 + x2

Eventually to obtain:

h(x1) = 2*x1 + x1/(2*x1 – 1)

Exercise 2
case A



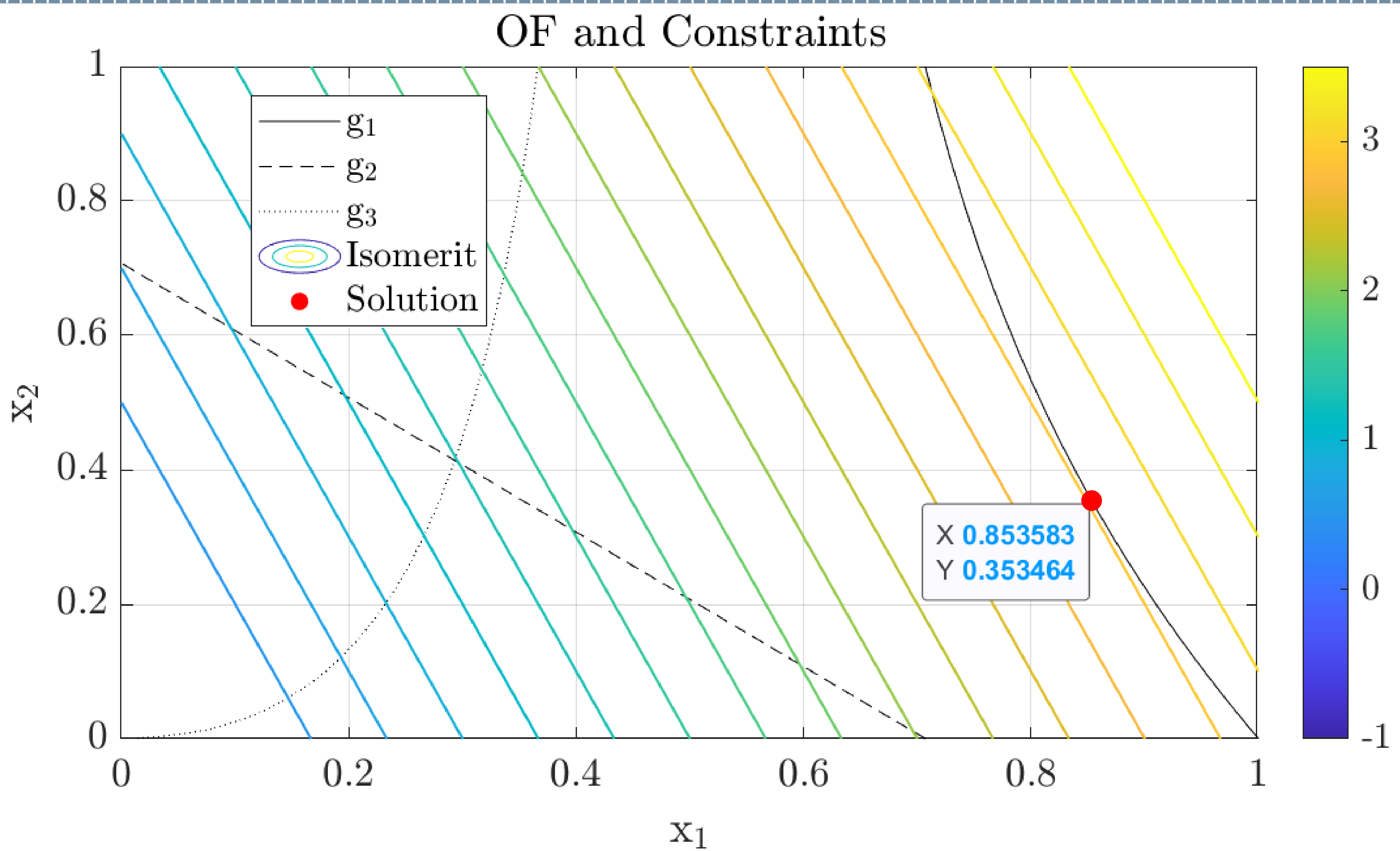
The function shows a minimum around 0.85.

Therefore, it must be the optimal value for x1.

That is, at x1 = 0.85 the objective function assumes its minimum value and satisfies all the constraints (the values of x1 we're looking at satisfy g1 = 0, while g2, g3 < 0).

The corresponding value of x2 can be found using the expression of the locus, which maps x2 into x1.

Exercise 2
case A



Exercise 2
Matlab codes – case A

Matlab code used to implement symbolic functions:

```
x1 = sym('x1', 'positive');
x2 = sym('x2', 'positive');
% objective function
f(x1, x2) = 3*x1 + x2;
% constraint 1
g1(x1, x2) = (2*x1+x2)/(2*x1*(x1+x2)) - 1;
% constraint 2
g2(x1, x2) = 1/(sqrt(2)*(x1+x2)) - 1;
% constraint 3
g3(x1, x2) = x2/(2*x1*(x1+x2)) - 1;
```

$$(\mathbb{SO})^{5a}_{\mathrm{nf}} \begin{cases} \min\limits_{x_1,x_2} \ 3x_1 + x_2 \\ \text{s.t.} \begin{cases} \dfrac{2x_1+x_2}{2x_1(x_1+x_2)} - 1 \leq 0 \\ \dfrac{1}{\sqrt{2}(x_1+x_2)} - 1 \leq 0 \\ \dfrac{x_2}{2x_1(x_1+x_2)} - 1 \leq 0 \\ x_1 > 0, \qquad x_2 \geq 0, \end{cases} \end{cases}$$

Matlab code used to plot 2D contours of the constraints:

```
% contour plot of the constraints
figure()
fcontour(g1, 'k--', [0 1], 'levellist', 0, 'displayname', 'g$_1$');
hold on
fcontour(g2, 'b--', [0 1], 'levellist', 0, 'displayname', 'g$_2$');
fcontour(g3, 'r--', [0 1], 'levellist', 0, 'displayname', 'g$_3$');
hold off
legend('box', 'off', 'interpreter', 'latex')
grid on
title('Constraints', 'interpreter', 'latex')
xlabel('x$_1$', 'interpreter', 'latex')
ylabel('x$_2$', 'interpreter', 'latex')
set(gca, 'fontsize', 15, 'ticklabelinterpreter', 'latex')
```

Exercise 2
Matlab codes – case A

Matlab code used to solve the problem:

```
% find x2(x1) such that g1 = 0
soluzione = solve(g1, x2, 'returnconditions', true);
locus(x1) = soluzione.x2;
% substitute the locus into f
f_cons(x1) = subs(f, x2, locus);
% plot
figure()
fplot(f_cons, [0.55 1])
xlabel('x_1')
title('Objective Function value on g_1 = 0 locus')
grid on
axis tight
set(gca, 'fontsize', 13)
% looking for the minimum
dfdx1 = diff(f_cons, x1);
x1_struc = solve(dfdx1, x1, 'returnconditions', true);
x1_sol_num = double(x1_struc.x1);
x2_sol_num = double(locus(x1_sol_num));
% the admissible solution is x1 = 0.8536 and x2 = 0.3536
x1_min = x1_sol_num(2);
x2_min = x2_sol_num(2);
```

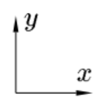
Exercise 2
case B

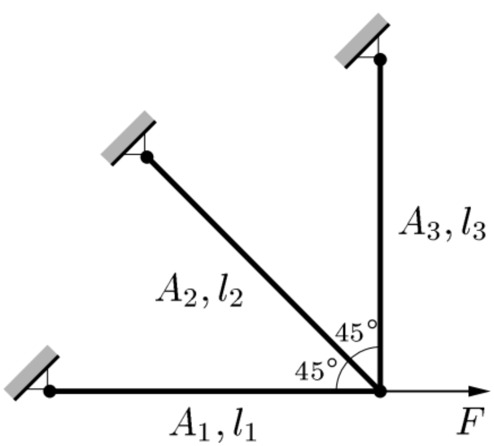
CASE B)

$\rho_1 = \rho_2 = \rho_3 = \rho_0, \ \sigma_1^{\max} = \sigma_3^{\max} = 2\sigma_0, \ \sigma_2^{\max} = \sigma_0.$

Using the same dimensionless variables as for the previous case, we may write the problem as

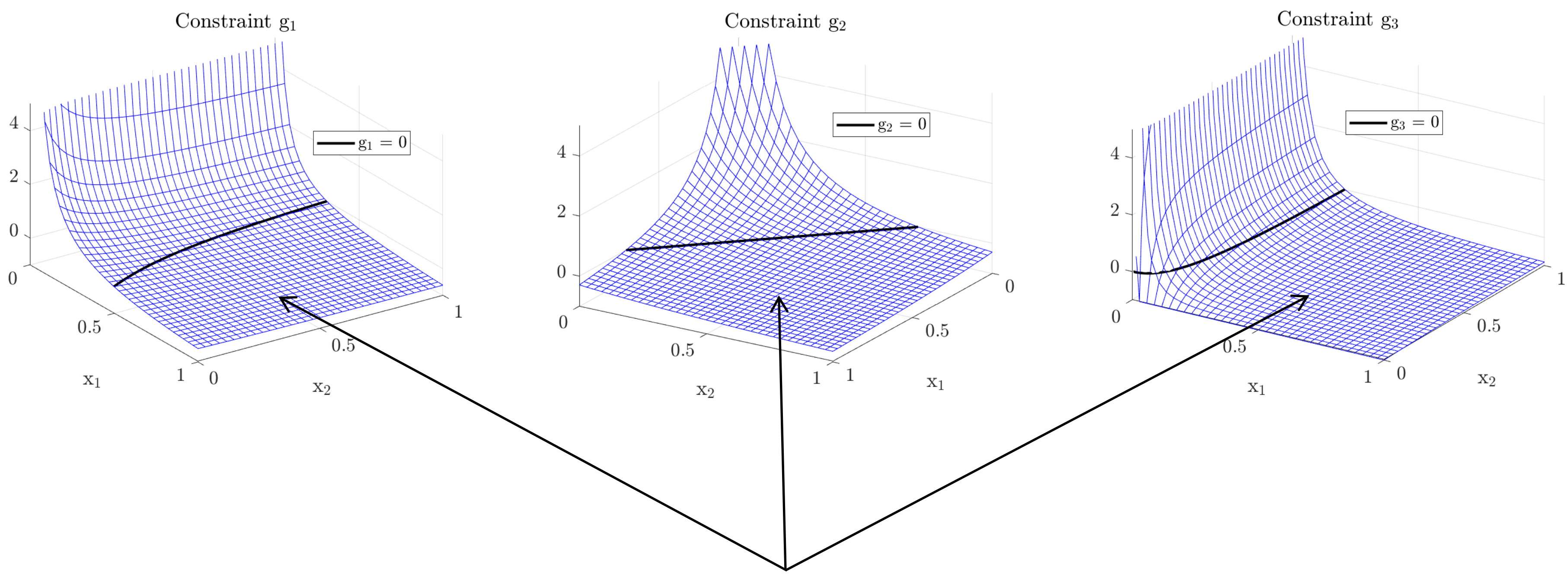
$$
(\mathbb{SO})^{5b}_{\mathrm{nf}} \quad \begin{cases} \min\limits_{x_1, x_2} \ 2x_1 + x_2 \\ \text{s.t.} \quad \begin{cases} \dfrac{2x_1 + x_2}{4x_1(x_1 + x_2)} - 1 \le 0 & & (\sigma_1) \\ \dfrac{1}{\sqrt{2}(x_1 + x_2)} - 1 \le 0 & \text{if } x_2 > 0 & (\sigma_2) \\ \dfrac{x_2}{4x_1(x_1 + x_2)} - 1 \le 0 & & (\sigma_3) \\ x_1 > 0, \qquad x_2 \ge 0, \end{cases} \end{cases}
$$





With the help of Matlab (cf. the codes provided you in the slides before) plot the constraints trend in the $x_1 - x_2$ cartesain plane. Overlap the objective function and try to identify (approximately) the solution to the problem.

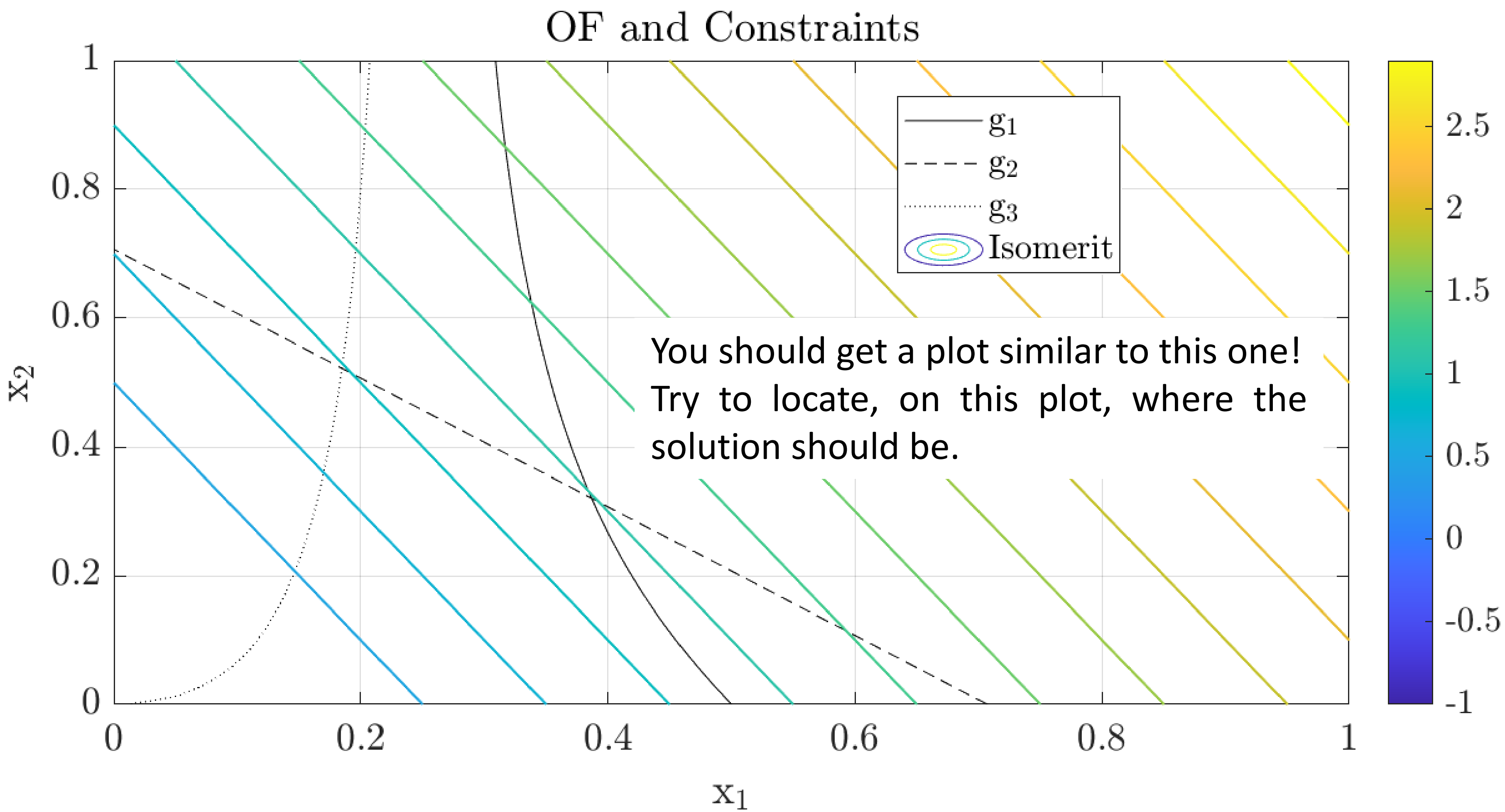
Exercise 2
case B



Feasible regions

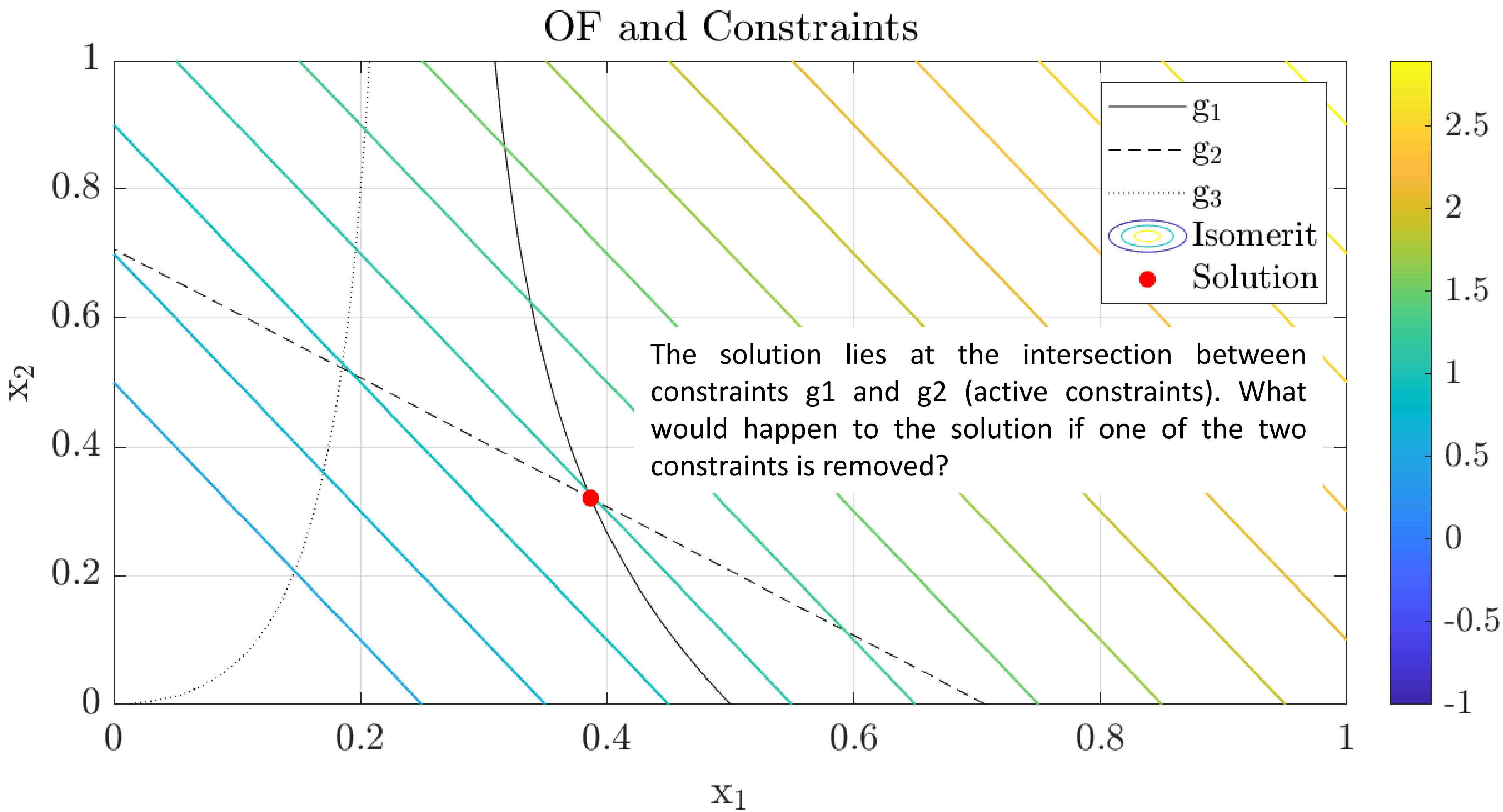
$g_i < 0$

Exercise 2
case B



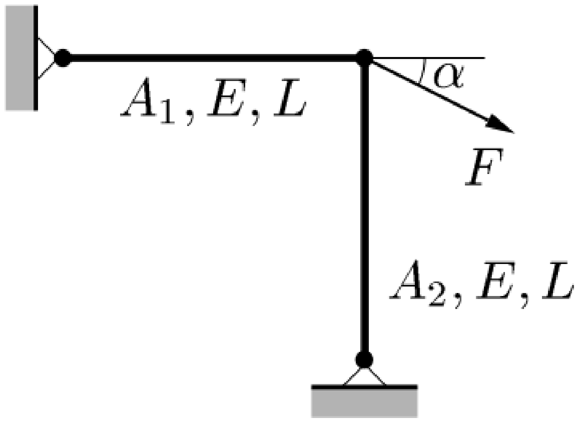
You should get a plot similar to this one! Try to locate, on this plot, where the solution should be.

Exercise 2
case B – solution



The solution lies at the intersection between constraints g1 and g2 (active constraints). What would happen to the solution if one of the two constraints is removed?

Exercise 1 bis



Consider a two-bar truss consisting of bars of length L and Young's modulus E. The force F > 0 is applied at an angle α = 45°.

The problem is **to find the circular cross-sectional areas A_1 and A_2 such that the stiffness of the structure is maximized under constraints on volume and yielding**.

$$
(\mathbb{SO}) \quad \begin{cases} \text{minimize } f(x, y) \text{ with respect to } x \text{ and } y \\ \text{subject to} \quad \begin{cases} \text{behavioral constraints on } y \\ \text{design constraints on } x \\ \text{equilibrium constraint.} \end{cases} \end{cases}
$$